

Rank Before You Merge: Training-Free Machine-Centric Visual Token Compression for Efficient Vision-Language Model Inference

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Abstract

Native spatial mergers already compress visual patches before a vision-language model (VLM) reaches its language model, but most token selectors rank the merger outputs. We treat the model’s merge topology and visual interface as compression constraints and introduce Rank-Before-Merge (RBM), a training-free rule that ranks complete native units before applying the unmodified merger. At 25% retention, RBM outperforms matched post-merger L2 selection on all nine text-dense full-split cells across Qwen3-VL, Qwen2.5-VL, and InternVL3, with gains of 11.0–38.4 percentage points and 235–432 OCRBench points. A merger-aware rank prior also improves FastV-k3 on all four locked Qwen3-VL probes at the same budget, although only TextVQA is significant and pure RBM remains stronger on OCRBench. Rank and token-survival analyses attribute the main effect to merger-induced saliency rewriting. RBM adds no learned parameters and increases measured throughput by 68% on Qwen3-VL at 25% retention.

Introduction

Modern VLMs can expose hundreds to tens of thousands of visual tokens to the language model. Reducing that sequence lowers attention and key-value-cache cost, so visual-token compression has become an important part of efficient multimodal inference. Merger-equipped models introduce a structural detail that current methods often treat as fixed plumbing: neighboring patch features are first compressed into native spatial units, and a later selector decides which merged tokens survive. If the merger changes saliency order, a precise top- k decision made after it cannot recover an answer-bearing unit that was ranked below the threshold.

We call a compressor *machine-centric* when its selection unit, budget, position handling, stream identity, and output interface follow the native computation graph of the deployed model. This term does not imply a new learned importance model. It distinguishes compression that respects the machine’s existing representation and topology from a selector applied to a convenient but already altered token stream. Figure 1 shows the resulting design: Rank-Before-Merge (RBM) evaluates complete 2×2 units on merger-input features, keeps a fixed per-image budget, and invokes the original merger only on the survivors.

Our first contribution is a representation-dependent formulation of visual token selection around a lossy native merger, including the condition under which pre- and post-merger decisions agree. Second, we introduce RBM as a training-free compressor with an explicit native-unit and interface contract. Across three VLMs and two

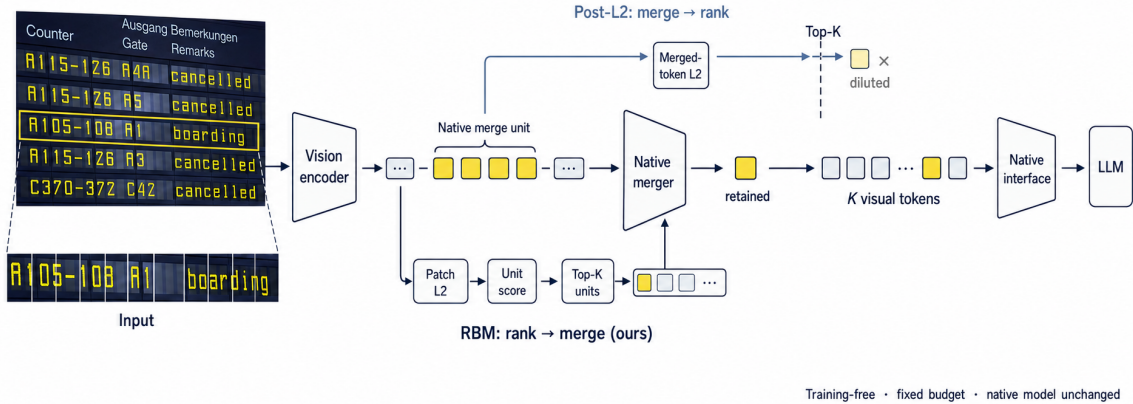


Figure 1: RBM follows the native merge topology. It ranks complete merger-input units, preserves a fixed per-image budget, and passes the selected units through the unmodified merger and language-model interface. The comparison changes selection order, not model weights.

merger designs, RBM preserves substantially more text-dense evidence than matched post-merger selection at equal token count. Third, we test whether the pre-merger signal complements a strong query-conditioned selector. A locked RankBridge probe reserves 20% of the final budget for pre-merger-ranked units and fills the remaining budget using FastV attention. It improves FastV-k3 on all four tested Qwen3-VL benchmarks by 0.5–3.2 percentage points, with a significant TextVQA gain. This establishes that pre-merger and query-conditioned ranks contain complementary signals. Finally, controlled rank, edge-energy, and ranking-swap analyses connect the accuracy gap to merger-induced saliency rewriting.

Related Work

Training-free VLM compression commonly prunes tokens before or inside the language model. FastV uses question-conditioned attention in an early LLM layer [1]; PyramidDrop progressively shortens the visual sequence [2]; FasterVLM and SparseVLM refine attention- and query-based selection [3, 4]. PruMerge and FitPrune combine selection with learned or fitted importance signals [5, 6], while VisionZip and AdaptMerge study token merging and the loss of fine visual detail [7, 8]. Recent systems combine importance with diversity or text guidance [9, 10, 11].

These methods mainly ask which token or feature should survive. Our controlled variable is the representation on which that decision is made. This distinction matters for Qwen-VL, whose learned PatchMerger concatenates four neighboring patch vectors and applies LayerNorm and a nonlinear MLP, and for InternVL3, which uses pixel-shuffle downsampling [12, 13, 14]. RBM neither replaces the native merger nor proposes another importance network. It changes the order of selection and aggregation while preserving the model’s visual interface.

Machine-Centric Compression

Selection around a native merger

Let $U = \{u_i\}_{i=1}^N$ be the native merge units and let M be the model’s per-unit merger. A representation-dependent scorer produces $a_i = s_{\text{pre}}(u_i)$ before merging and $b_i = s_{\text{post}}(M(u_i))$ after merging. At budget k , the two decisions are

$$K_{\text{pre}} = \text{TopK}_k\{a_i\}, \quad K_{\text{post}} = \text{TopK}_k\{b_i\}. \quad (1)$$

With deterministic tie breaking, they are equivalent exactly when $K_{\text{pre}} = K_{\text{post}}$. A sufficient condition for agreement at every budget is $b_i = g(a_i)$ for a strictly increasing g . We measure the violation at budget k by

$$D_k = 1 - \frac{|K_{\text{pre}} \cap K_{\text{post}}|}{|K_{\text{pre}} \cup K_{\text{post}}|}. \quad (2)$$

For a fixed unit set, selection and a per-unit merger are compatible. The non-equivalence arises because the merger changes the ranking that determines the set.

Rank-Before-Merge

For every image, RBM groups ViT patch features into the model’s native 2×2 units. For a unit u with input patch vectors $h_{u,1:4}$, it uses the query-blind control score

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2. \quad (3)$$

It retains $k = \max(1, \text{round}(\kappa N))$ units and applies the native merger to those units. The method contract has four parts: selection uses complete native units; every image receives its exact top- k budget; the same unit identities propagate through main and deepstack streams; and the native merger, positional encoding, visual interface, and downstream LLM are unchanged. L2 scoring costs $O(Nd)$ for d -dimensional features and top- k selection costs $O(N \log k)$ with a heap. RBM adds no parameters and does not require question tokens or an LLM-attention pass.

Because Qwen’s merger is applied independently per unit, supplying a fixed kept set to the original operator produces the same merged representations. This is byte-identical in the verified Qwen3-VL path. The invariant lets us separate a changed ranking from a changed forward operator in the swap control below.

A merger-aware prior for FastV

FastV ranks the full visual-token candidate set with question-conditioned attention at LLM layer $K = 3$. RankBridge tests whether the merger-input order contains information that this later signal misses. At retention $\kappa = 0.25$, it protects $q = \text{round}(\rho k)$ positions selected by the RBM pre-rank and fills the remaining $k - q$ positions by FastV rank. A development gate over $\rho \in \{0.1, 0.2, 0.3\}$ selected $\rho = 0.2$; the value was then locked for four $n = 200$ benchmarks. At $\rho = 0$, the implementation is bit-identical to FastV; at $\rho = 1$, its kept set is identical to RBM. All visual tokens remain available until layer K , so the pre-rank does not starve FastV’s candidate set.

Kept visual units (blue) per selector — two OCR cases

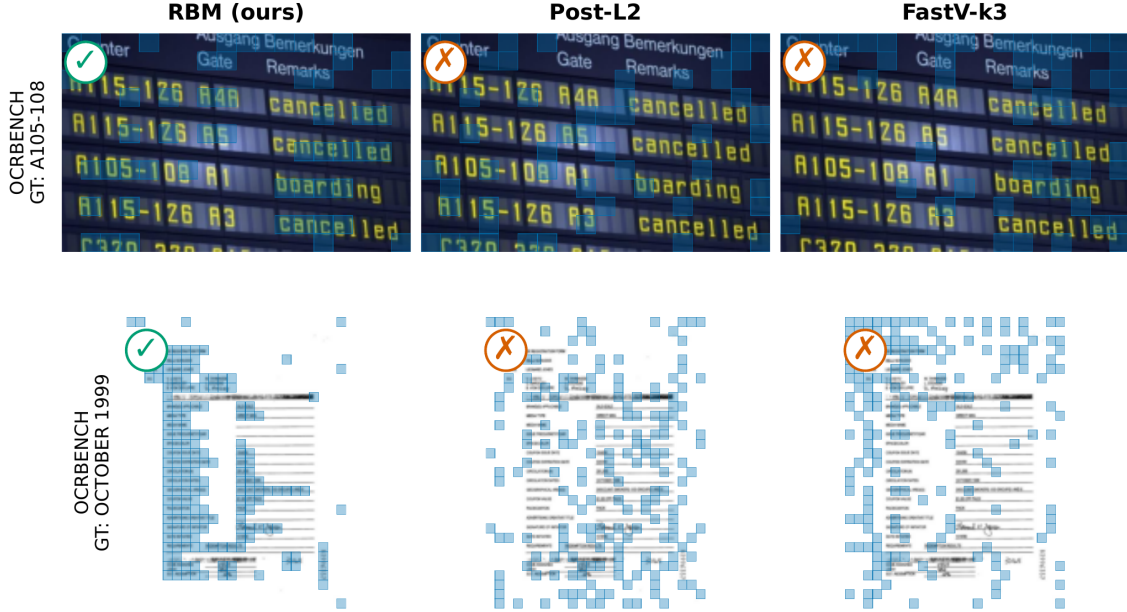


Figure 2: Audited OCRBench examples at 25% retention. RBM, Post-L2, and FastV-k3 preserve different spatial evidence. The examples illustrate selection behavior; aggregate results are reported separately.

Experimental Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B. The Qwen models use a learned PatchMerger and InternVL3 uses pixel-shuffle downsampling. All runs use bf16 weights, greedy decoding, and family-correct native positions. RBM, no-compression, and Post-L2 cells use vLLM 0.19 V1 [15]; FastV and RankBridge use a HuggingFace eager-attention harness because FastV has no vLLM path. Zero-drop anchors agree across the two harnesses, but we report throughput only within vLLM.

The benchmarks are TextVQA with official VQA accuracy [16], DocVQA with ANLS [17], OCRBench with its /1000 score [18], and GQA with normalized exact match [19]. Full splits contain 5000, 5349, 1000, and 12578 items. Scorers are ports of the lmms-eval implementations [20]. Pre/post comparisons pair identical samples and per-image token counts. The nine prespecified text-dense comparisons at 25% retention use 20,000-resample paired-bootstrap intervals and paired sign-flip tests with Holm correction; every corrected p is at most 4.5×10^{-4} . FastV and RankBridge comparisons are secondary locked probes and use paired tests on $n = 200$.

Compression Results

Matched pre- and post-merger selection

Table 1 isolates selection stage within the same query-blind L2 family. At 25% retention, RBM outperforms Post-L2 in all nine text-dense cells. The Qwen3-VL gains

Table 1: Official full-split results. RBM and Post-L2 use the same per-image visual-token count. Δ is RBM minus Post-L2 at 25% retention. OCRBench uses points; other rows use percentage points.

Model	Benchmark	None	RBM	Post-L2	Δ
Qwen3-VL	TextVQA	.844	.605	.222	+38.4
	DocVQA	.956	.481	.238	+24.3
	OCRBench	760	547	184	+363
	GQA	.616	.449	.477	−2.8
Qwen2.5-VL	TextVQA	.862	.702	.442	+26.1
	DocVQA	.949	.636	.526	+11.0
	OCRBench	817	480	182	+298
	GQA	.604	.559	.585	−2.6
InternVL3	TextVQA	.8338	.7890	.4148	+37.4
	DocVQA	.9221	.7284	.3820	+34.6
	OCRBench	852	753	321	+432
	GQA	.6293	.5993	.6031	−0.4

are 38.4 pp on TextVQA, 24.3 pp on DocVQA, and 363 OCRBench points; Qwen2.5-VL gains 26.1 pp, 11.0 pp, and 298 points. The direction replicates with InternVL3’s different merger: 37.4 pp, 34.6 pp, and 432 points. Deeper 12.5% retention widens the gap on the reported text-dense cells. RBM is not lossless relative to no compression, especially on InternVL3 DocVQA and OCRBench. The evidence establishes improved retention over a matched post-merger decision.

GQA gives the boundary of the result. The operational Qwen rankings favor Post-L2 by 2.6–2.8 pp, while InternVL3 differs by only −0.4 pp with a 95% interval crossing zero. A full-split Qwen3-VL matched-boundary control ranks the pre arm at the main merger’s own input. It confirms before-merger gains of 27.68 pp on TextVQA, 4.59 pp on DocVQA, and 235 OCRBench points, but favors Post-L2 by 5.64 pp on GQA (95% CI $[-6.42, -4.86]$). The result is a workload-conditioned stage effect, not a universal preference for early selection.

FastV and the merger-aware prior

Table 2 reports the locked RankBridge probe at the same 25% budget. The merger-aware prior improves FastV-k3 on every benchmark. TextVQA gains 3.2 pp with paired $z = 2.11$; the 0.5–1.6 pp gains on the other three benchmarks are not significant at $n = 200$. Pure RBM remains 11.6 pp stronger than RankBridge on OCRBench, so the hybrid fails a preregistered requirement to stay within 1 pp of the stronger parent on every benchmark. This negative gate is informative: pre-merger rank contributes information beyond query attention, but a single global mixture cannot replace the workload boundary.

The separate same-scope baseline campaign also shows why the pure methods remain useful. On full OCRBench, RBM exceeds FastV by 14.6 pp on Qwen3-VL and 8.7 pp on Qwen2.5-VL. FastV is stronger on Qwen3-VL TextVQA and GQA and on Qwen2.5-VL TextVQA. Query-conditioned selection and topology-preserving selection therefore occupy different regimes.

Table 2: Locked Qwen3-VL probe at 25% retention ($n = 200$). RankBridge protects 20% of its final budget with the pre-merger rank and fills the remainder with FastV-k3 attention.

Benchmark	RBM	FastV-k3	RankBridge	RB-F
TextVQA	.5967	.7633	.7950	+3.2
DocVQA	.4239	.5863	.6023	+1.6
OCRBench	.5801	.4586	.4641	+0.6
GQA	.4150	.5050	.5100	+0.5

Why the Merger Changes Selection

Figure 3 summarizes three diagnostics on Qwen3-VL. First, pre- and post-merger unit ranks are weakly related. Spearman correlation is 0.137 on DocVQA, 0.332 on TextVQA, and 0.360 on GQA; top-25% kept-set Jaccard is 0.180, 0.243, and 0.278. The native merger therefore changes the actual top- k decision, not only score scale.

Second, the changed order is structured. With rank shift defined as post rank minus pre rank, its correlation with Sobel edge energy is +0.439 on DocVQA, +0.155 on TextVQA, and +0.036 on GQA. On DocVQA, units kept by RBM but dropped by Post-L2 have mean edge energy 0.641, compared with 0.124 in the reverse group; 92% lie above the per-image median, compared with 35%. This supports text-stroke demotion as a mechanism-level association. It does not establish the merger’s training objective as the cause.

Third, a ranking-swap control holds the post-merger forward path fixed but uses the pre ranking. On Qwen3-VL, it selects exactly the RBM set (Jaccard = 1.000) and recovers pre accuracy, while the retained merged outputs are byte-identical. The measured Qwen3-VL gap is therefore attributable to the selection ranking rather than a different operation on the kept units.

What the native merger changes

per-image distributions | 25% retained

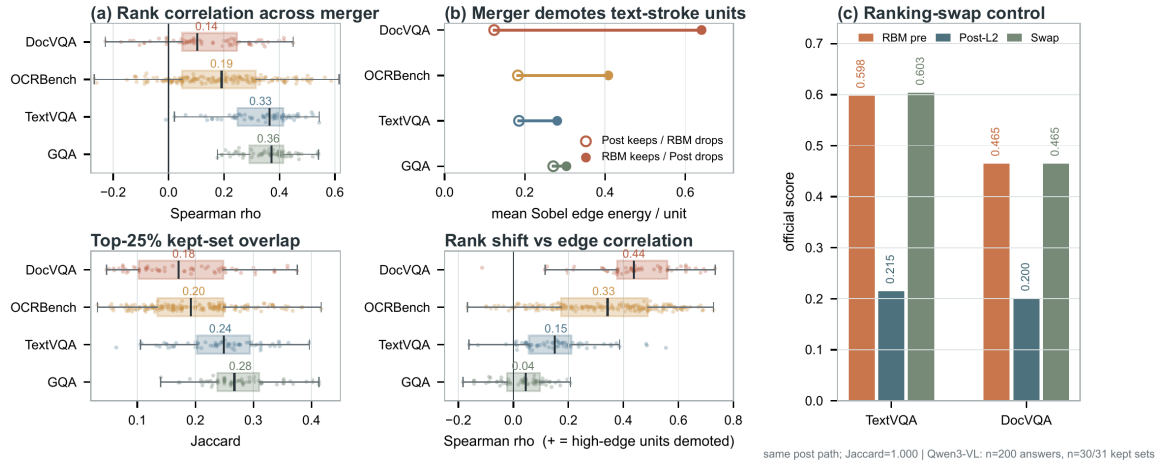
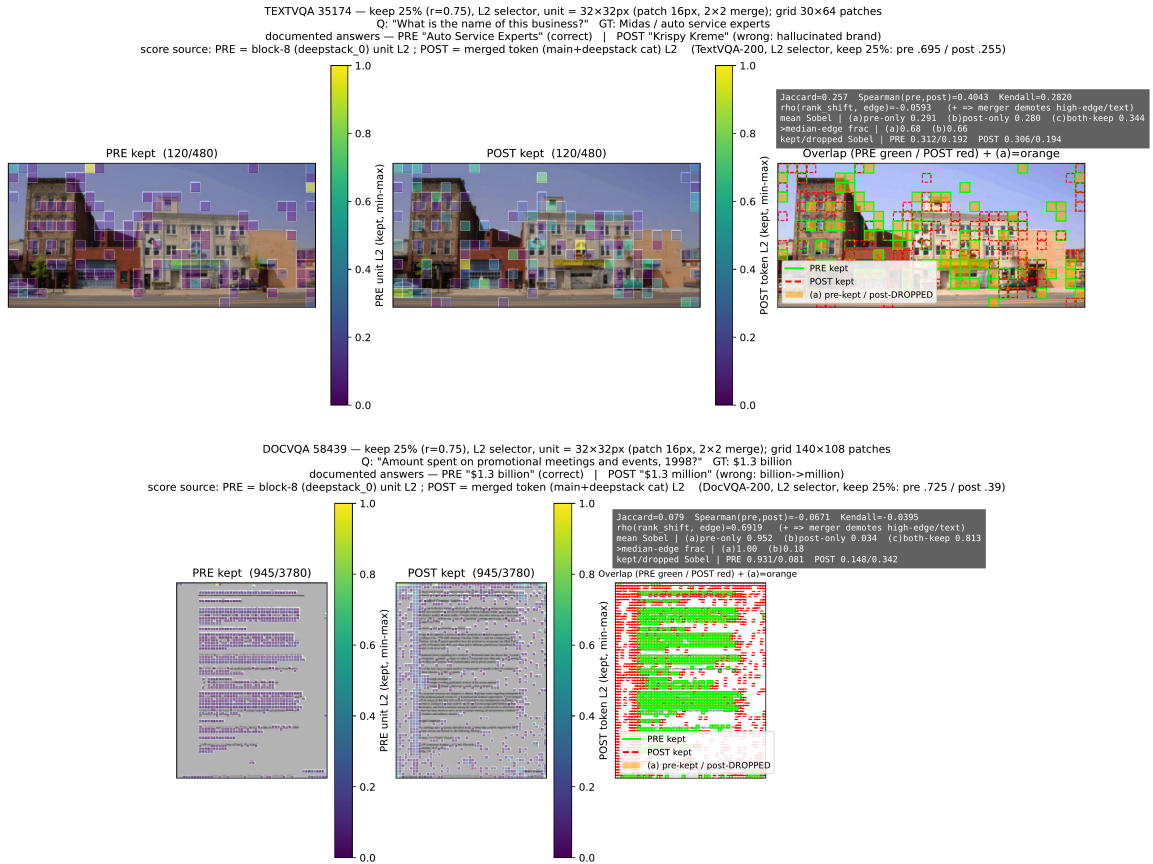


Figure 3: The native merger rewrites selection saliency. Rank agreement is low, text-stroke-rich units are demoted, and a ranking-swap control restores RBM accuracy with the same kept set.



Efficiency and Scope

Efficiency is measured in vLLM 0.19 V1 on one A40 GPU. Across 75%, 50%, and 25% retention, RBM and Post-L2 throughput is 4.61/4.52, 5.39/5.54, and 6.91/6.89 requests/s. Five-repeat measurements remain within 3% with no consistent stage difference. Relative to no compression, RBM raises throughput by 12%, 31%, and 68%; at 25% retention, latency falls from 0.36 to 0.23 seconds. InternVL3 independently yields 1.8–2.5 $\times$ throughput at 25% retention. RBM does not reduce ViT computation before its scoring tap; its gain comes from shortening the sequence presented to the native merger and language model.

The byte-exact ranking attribution is established on Qwen3-VL. Qwen2.5-VL preserves kept-set identity but has an ordering confound, and InternVL3 offers accuracy-level replication and a bounded swap result. Models without an identifiable native merge unit are outside the method’s scope. FastV and RankBridge accuracy uses a different eager harness from the vLLM efficiency study. Several secondary FastV and mechanism cells use $n = 200$, as marked, and only the TextVQA RankBridge gain is significant. Pixel caps differ across some model families, so absolute cross-model scores should not be compared. The current result supports a merger-aware machine-centric design rule for text-dense compression, not a universal ranking stage or selector.

Conclusion

Visual-token selection around a native merger is an operator-order problem. RBM resolves it by selecting complete native units on merger-input features and applying the unchanged model only to the survivors. Under equal token budgets, this training-free design preserves substantially more text-dense evidence than matched Post-L2 across three VLMs and two merger designs while delivering the expected inference speedup. A locked merger-aware rank prior also improves FastV-k3 on all four tested Qwen3-VL probes, indicating that pre-merger and query-conditioned ranks contribute complementary information. The mechanism results show why: the native merger changes saliency order and preferentially demotes text-stroke units. Compression methods for merger-equipped VLMs should therefore treat native topology and selection stage as first-class design constraints.

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